OPERATING SYSTEMS

UE25CS242A

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File Management

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OPERATING SYSTEMS

System and Network Threats

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Slides Credits for all PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

System Threats



- System threats are attacks that target the system services, resources, or operating environment.
- The goal may be to:
  - Make system resources unavailable
  - Consume excessive CPU, memory, or storage
  - Gain unauthorized access
  - Exploit weaknesses in system services
  - Disrupt normal system operation

Common System Threats

1. Denial-of-Service (DoS) - Attempts to make a system or service unavailable to legitimate users.

2. Port Scanning - Systematically checks network ports to identify available services and potential entry points.

Denial-of-Service (DoS)



A Denial-of-Service (DoS) attack attempts to make a system or network service unavailable to legitimate users.

The attacker overwhelms a system by consuming its:

- CPU resources
- Memory
- Network bandwidth
- Connection capacity
- Other system resources

Port Scanning



- Port scanning is a technique used to examine a computer or network system to identify open network ports and the services associated with them.
- Network services listen on specific port numbers for incoming connections.
- Attackers may use port scanning to discover potential entry points and vulnerable services.
- Administrators also use port scanning for network security auditing and troubleshooting.

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Network Threats



- Network threats are attacks that exploit vulnerabilities in network communication, protocols, or connected systems.
- Attackers may attempt to:
  a. Intercept network communication
  b. Steal sensitive information
  c. Impersonate legitimate systems or users
  d. Modify transmitted data
  e. Disrupt network services

Common Network Threats

1. IP Spoofing - Attacker sends packets using a forged source IP address.

2. Man-in-the-Middle (MITM) - Attacker secretly positions themselves between two communicating parties and may observe or modify communication.

3. Packet Sniffing - Attacker captures and analyzes network packets to obtain information.

IP Spoofing & Man-in-the-Middle (MITM) Attack



- IP spoofing is a technique in which an attacker forges the source IP address of network packets.
- The packet appears to come from a trusted or different system.
- The actual sender's IP address is hidden or replaced with a false address.

- A Man-in-the-Middle (MITM) attack occurs when an attacker secretly positions themselves between two communicating parties.
- The attacker can potentially intercept, read, or modify the communication.
- The sender and receiver may believe they are communicating directly with each other

Packet Sniffing



- Packet sniffing is the process of capturing and analyzing network packets as they travel through a network.
- A packet sniffer is a software or hardware tool used to inspect network traffic.
- Packet sniffing can be:
  - **Legitimate** — network monitoring, troubleshooting, and security analysis.
  - **Malicious** — capturing sensitive information without authorization.

Information That May Be Exposed

- Usernames and passwords
- Network addresses
- Communication data
- Session information
- Other unencrypted information

Network Security & Protection Mechanisms



How Can Systems Be Protected?

1. Firewalls

- Control incoming and outgoing network traffic based on security rules.
- Can block unauthorized connections and services.

2. Encryption

- Converts readable data into **unreadable ciphertext**.
- Protects information during network transmission.
- Helps prevent attackers from understanding captured data.

3. Authentication

- Verifies the **identity of users or systems**.
- Examples: passwords, certificates, and multi-factor authentication.



THANK YOU

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